

Report

Topic content (NLP05): *A Comparative Study of PEGASUS, BART, and T5 for Text Summarization Across Diverse Datasets*

1. Study Scope and Methodology

The study provides a comprehensive comparative evaluation of three leading transformer-based models—PEGASUS, BART, and two variants of T5 (T5-Small and T5-Base)—for abstractive text summarization. These models are built upon the Transformer architecture, utilizing self-attention mechanisms to deliver fluent and context-aware summaries.

1.1. Model Architectures

- PEGASUS: Optimized for summarization via a unique pre-training strategy called gap-sentence generation (GSG), forcing the model to reconstruct strategically removed sentences. The PEGASUS-CNN/DailyMail variant was used.
- BART: Functions as a denoising autoencoder, combining a bidirectional BERT encoder and an autoregressive GPT decoder. Its pre-training uses five noise strategies, including sentence permutation and text infilling, making it robust at generating coherent text from corrupted sequences. The BART LARGE version was employed.
- T5 Variants: Unifies all NLP tasks into a text-to-text problem. T5-Small (~60 M parameters) was included for lightweight efficiency, and T5-Base (~220 M parameters) was included to evaluate a moderate-resource scenario.

1.2. Datasets and Evaluation

The evaluation utilized a unified framework addressing accuracy, computational efficiency, and domain adaptability. Models were fine-tuned using filtered and preprocessed subsets of the data:

1. CNN/DailyMail: Long-form news articles, emphasizing context retention.
2. Xsum: Extreme summarization, requiring single-sentence abstraction.
3. Samsum: Conversational dialogues, requiring distillation of informal language.

Performance was measured quantitatively using ROUGE-1, ROUGE-2, and ROUGE-L scores, alongside efficiency metrics like training time. A crucial component was the

qualitative assessment by human experts, rating summaries on fluency, relevance, and conciseness, and identifying errors like hallucination and factual errors.

2. Key Findings and Practical Insights

The results demonstrated that no single model is universally optimal; the best choice depends heavily on the task and resource constraints.

Dataset/Task	Top Model Performance (Phase 1 ROUGE-1)	Key Strengths & Context
CNN/DailyMail (Long-Form News)	PEGASUS (35.49%)	Highest ROUGE scores and superior factual retention, attributed to its GSG pre-training alignment with news data. Recommended for high-stakes applications where factual accuracy is paramount.
Xsum (Extreme Abstraction)	BART LARGE (41.40%)	Excelled in abstractive tasks due to its denoising autoencoder objective, resulting in greater coherence.
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2.1. Efficiency and Model Scaling

- T5-Small consistently yielded the lowest ROUGE, but offered substantial advantages in computational efficiency and resource-constrained settings (1 h 05 m training time on CNN/DailyMail).
- T5-Base provided a practical middle ground, narrowing the performance gap with larger models like BART and PEGASUS while maintaining moderate efficiency advantages compared to them. Scaling up from T5-Small to T5-Base significantly improved output quality.

2.2. Qualitative Reliability

Qualitative analysis revealed that while PEGASUS leads in factual retention on news, T5-Small struggled the most, recording the highest factual errors (9% on CNN/DailyMail) and context loss (10%). All models were prone to hallucination (inventing non-existent facts), particularly on the extreme abstraction task Xsum (PEGASUS 12%, BART 10%, T5-Small 15%).

3. Conclusion

The study concludes that model selection must be task-driven, balancing accuracy, linguistic style, and computational cost.

For instance, PEGASUS is best for information-dense text, BART for fluency in conversational contexts, and T5-Small for low-resource environments where speed outweighs maximum accuracy.

Different Transformer-based summarization models - PEGASUS, BART, T5-Small, and T5-Base - perform differently depending on the dataset and task.

PEGASUS is best for factual long-form news; BART is strongest for highly abstractive tasks like Xsum; T5-Small is weakest but most efficient; T5-Base offers a good balance. All models suffer from hallucination, especially on extreme summarization.

This research shows the importance of:

- Matching model choice to the task
- Understanding scaling trade-offs between accuracy and efficiency
- Evaluating with both ROUGE and human judgment
- Recognizing domain-specific challenges in summarization

To develop this work into a full graduate thesis, I can add:

- A novel component (new model, retrieval augmentation, RL-based improvement)
- Stronger evaluation metrics (BERTScore, factuality metrics)
- A new dataset domain (legal, medical, financial)
- A deep analysis of hallucination causes and mitigation
- A hybrid extraction–abstraction pipeline
- A practical application system or dashboard